CAPSTONE PROJECT
Global Economic Analysis using World Bank Data

Project Objective

To analyze global economic performance using World Bank data by examining GDP trends, population dynamics, and unemployment patterns. The goal is to convert raw data into actionable business insights using data analysis and visualization techniques.

Task 1: Data Collection (Web Scraping & API)

Techniques Used:

- Used **World Bank API** to fetch real-time economic data
- Extracted structured JSON data using **Python requests library.**
- Converted 1000+ data records raw API response into tabular format using **pandas Data Frame**
- Automated data retrieval for multiple countries and years

```
[7]: ## Coverting Data fame to csv file
     df.to_csv("world_bank_data.csv", index=False)

[8]: df_check = pd.read_csv("world_bank_data.csv")

     print(df_check.head(10))

                        country  year           gdp   population  inflation  \
0  Africa Eastern and Southern  2025           NaN          NaN        NaN
1  Africa Eastern and Southern  2024  1.242694e+12  769280888.0   4.770857
2  Africa Eastern and Southern  2023  1.179359e+12  750491370.0   7.399186
3  Africa Eastern and Southern  2022  1.228968e+12  731821393.0  10.883478
4  Africa Eastern and Southern  2021  1.114145e+12  713090928.0   6.824727
5  Africa Eastern and Southern  2020  9.386076e+11  694446100.0   5.191629
6  Africa Eastern and Southern  2019  1.018715e+12  675950189.0   4.102851
7  Africa Eastern and Southern  2018  1.020956e+12  657801085.0   4.689806
8  Africa Eastern and Southern  2017  9.780765e+11  640058741.0   6.221375
9  Africa Eastern and Southern  2016  8.318681e+11  623369401.0   6.446877

   unemployment  gdp_growth
0      7.539234         NaN
1      7.571697    2.763839
2      7.676548    1.931160
3      7.869470    3.722717
4      8.407412    4.578772
5      7.974302   -2.817572
6      7.459106    2.030077
7      7.244843    2.705194
8      7.214915    2.677524
9      7.074429    2.130117

[10]: df.shape

[10]: (2000, 7)
```

Task 2: Data Cleaning & Preprocessing

Techniques Used:

- Data cleaning done using python library Pandas.
- Handled missing values using forward fill / interpolation
- Converted columns into proper data types (int, float)
- Removed duplicates and inconsistent entries
- Created new features:
  - GDP per capita
  - GDP growth (%)
  - Population growth (%)
- Used pd.cut() to categorize GDP into Low, Medium, High

[61]:

	country	year	gdp	population	unemployment	gdp_growth	gdp_per_capita	growth_category	decade
0	Africa Eastern and Southern	1960	2.420569e+10	130075728.0	6.064025	4.018631	186.089204	Medium	1960
1	Africa Eastern and Southern	1961	2.495889e+10	133534923.0	6.064025	0.418937	186.909053	Low	1960
2	Africa Eastern and Southern	1962	2.707323e+10	137171659.0	6.064025	7.937038	197.367547	High	1960
3	Africa Eastern and Southern	1963	3.176914e+10	140945536.0	6.064025	5.623764	225.400079	High	1960
4	Africa Eastern and Southern	1964	3.027955e+10	144904094.0	6.064025	4.649241	208.962717	Medium	1960
5	Africa Eastern and Southern	1965	3.380618e+10	149033472.0	6.064025	5.138168	226.836135	High	1960
6	Africa Eastern and Southern	1966	3.692725e+10	153281203.0	6.064025	4.827421	240.911794	Medium	1960
7	Africa Eastern and Southern	1967	3.844416e+10	157704381.0	6.064025	5.338839	243.773593	High	1960
8	Africa Eastern and Southern	1968	4.174200e+10	162329396.0	6.064025	4.156334	257.143823	Medium	1960
9	Africa Eastern and Southern	1969	4.704852e+10	167088245.0	6.064025	5.121587	281.578894	High	1960

[60]:
```
df.to_csv("cleaned_world_bank_data.csv", index=False)
```

[38]:
```
df = df.reset_index(drop=True)
```

[43]:
```
df.shape
```

[43]:
```
(1876, 6)
```

[44]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1876 entries, 0 to 1875
Data columns (total 6 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   country       1876 non-null   object
 1   year          1876 non-null   int64
 2   gdp           1876 non-null   float64
 3   population    1876 non-null   float64
 4   unemployment  1876 non-null   float64
 5   gdp_growth    1876 non-null   float64
dtypes: float64(4), int64(1), object(1)
memory usage: 88.1+ KB
```

"After cleaning,there were no missing values, verified data types, and validated statistical summaries to confirm data quality. The dataset contained around 1800+ records across multiple countries and years, making it suitable for analysis."

Key Insights from Cleaning:

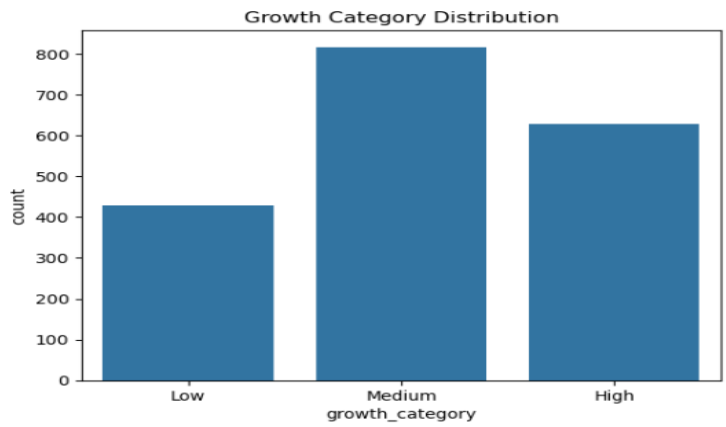
- Initial dataset had missing values in growth-related fields
- First-year records had NaN due to percentage change calculation
- Data normalization improved consistency for analysis

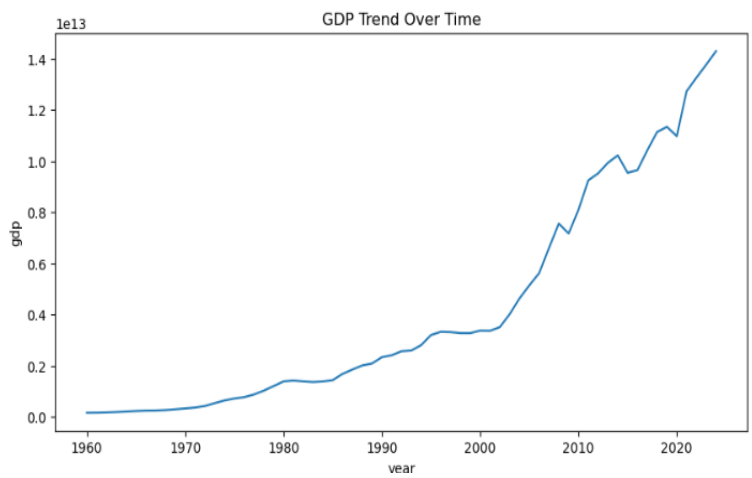
Task 3: Exploratory Data Analysis (EDA)

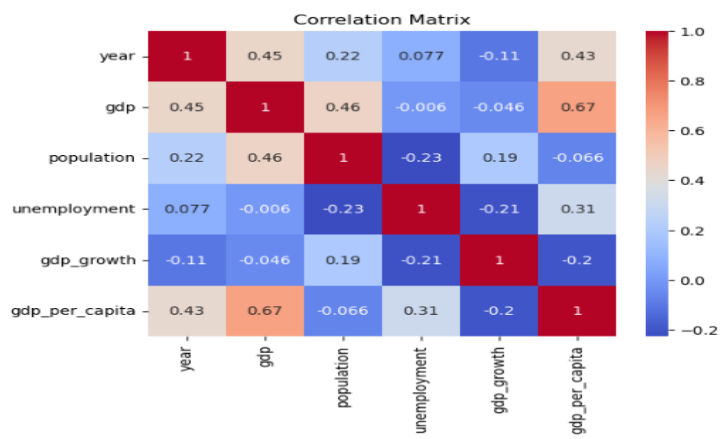
Techniques Used:

- Time-series analysis using line plots (Matplotlib/Seaborn)

- Aggregation using groupby() for country-wise insights
- Distribution analysis using categorical segmentation
- Trend analysis across decades








```
<Figure size 800x600 with 0 Axes>
```

Key Insights

- GDP Growth Trend:
  - Global GDP shows exponential growth, especially after 2000
  - Indicates acceleration due to globalization and industrialization
- Population Trend:
  - Population increases steadily over time
  - Suggests continuous pressure on resources and infrastructure
- Unemployment Trend:
  - Relatively stable with short-term fluctuations
  - Indicates resilience in global job markets
- Growth Category Insight:
  - Majority of countries fall under Medium growth category
  - Few countries dominate High GDP contribution
- Correlation Heatmap:
  - GDP and Population show strong positive correlation which indicates that countries with larger populations tend to have higher total GDP
  - GDP and GDP per capita show moderate correlation which suggests that economic output does not always directly translate to individual income
  - GDP growth and Population growth have weak correlation which indicates that economic growth is not solely driven by population increase

Task 4: Model Building

Techniques Used:

- Created derived metrics:
  - GDP growth rate
  - Population growth rate
  - GDP per capita
- Categorized economic performance using binning (pd.cut)
- Prepared dataset for dashboard consumption

- Used feature engineering to convert raw data into decision-ready variables

[102]:
```
## 7. Evaluating new model
from sklearn.metrics import mean_squared_error, r2_score

mse_rf = mean_squared_error(y_test, y_pred_rf)
r2_rf = r2_score(y_test, y_pred_rf)

print("Random Forest MSE:", mse_rf)
print("Random Forest R2:", r2_rf)
```

```
Random Forest MSE: 7.426578400367763
Random Forest R2: 0.24844917113456944
```

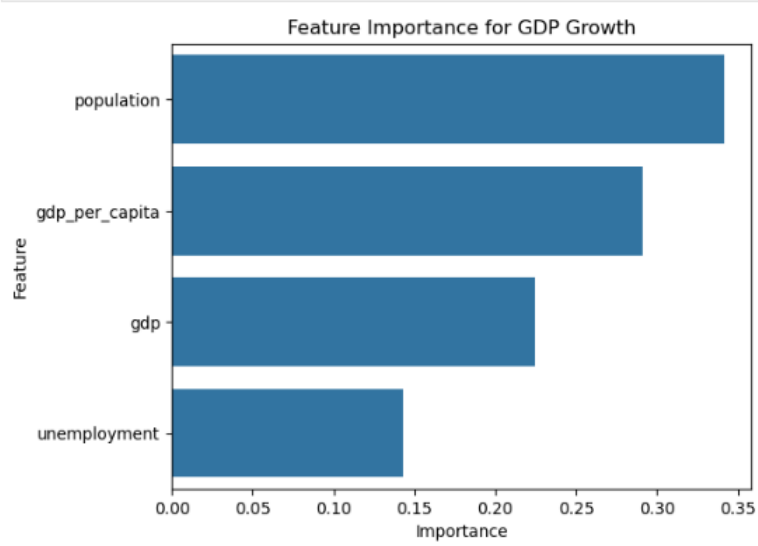
[107]:
```
## Feature Importance
## Shows which factor impacts GDP growth most

importance = pd.DataFrame({
    'Feature': X.columns,
    'Importance': rf_model.feature_importances_
}).sort_values(by='Importance', ascending=False)

print(importance)
```

```
          Feature  Importance
1      population    0.341321
3  gdp_per_capita    0.291244
0             gdp    0.224418
2    unemployment    0.143017
```

```
sns.barplot(x='Importance', y='Feature', data=importance)
plt.title("Feature Importance for GDP Growth")
plt.show()
```



Key Insights from Model / Features:

- GDP per capita reveals true economic strength
  - Large population ≠ strong economy
- Population growth vs GDP mismatch
  - Some regions show high population growth but low GDP growth
- Economic inequality visible
  - Few regions contribute majority of global GDP
- Growth categorization simplifies analysis
  - Helps in quick comparison across countries

- Feature engineering transformed raw economic data into meaningful indicators, enabling better visualization, comparison, and business decision-making in the dashboard.

Business Recommendations

1. **Population Optimization:**
   - Focus on skill development and workforce training to convert population into productive human capital rather than idle workforce.
   - Improve employment generation capacity by aligning education with industry demand to reduce mismatch in skills.
   - Optimize resource allocation in high-population regions to ensure infrastructure, jobs, and economic participation scale together.

2. **Improve GDP per Capita**:
   - Invest in high-value industries (technology, services, manufacturing) to increase output per individual rather than just total output.
   - Enhance education quality and digital skills to improve workforce efficiency and earning potential.
   - Strengthen income growth strategies (better wages, job quality) to improve overall economic performance and consumption capacity.

3. **Strengthen GDP:**
   - Focus on diversifying economic sectors to reduce dependency on limited industries and ensure stable growth.
   - Encourage exports and trade expansion to increase total economic output and global competitiveness.
   - Develop long-term economic planning to sustain GDP growth rather than short-term spikes.

4. **Reduce Unemployment:**
   - Create job opportunities through startups, MSMEs, and industrial expansion to absorb workforce efficiently.
   - Encourage entrepreneurship and self-employment models to reduce dependency on limited job sectors.
   - Monitor unemployment trends regularly to take early corrective actions before growth is impacted.

Task 5: Interactive Dashboard (Power BI)

Techniques Used in Dashboard Development

1. **Data Modeling & Preparation**
   - Imported cleaned dataset from Python into Power BI
   - Ensured correct data types (numeric, categorical, date)
   - Created calculated measures using DAX:
     - Total GDP = SUM(gdp)
     - Avg GDP per capita = AVERAGE(gdp_per_capita)
     - Avg Unemployment % = AVERAGE(unemployment)
   - Used pre-engineered features (GDP growth, population growth, categories) for better analysis

2. **KPI Design**
   - Built card visuals to highlight key indicators:
     - Total GDP → overall economic size

    - Average GDP per capita → individual economic strength
    - Total Population → scale of economy
    - Unemployment rate → labor market health
- Applied number formatting (Trillions, Thousands, %) for readability
- Used consistent styling and alignment to improve visual hierarchy

3. Time-Series Trend Analysis

- Created line charts:
    - GDP trend over time
    - Population growth trend
    - Unemployment trend
- Used continuous axis (year) for smooth trend visualization
- Enabled tooltips for detailed yearly insights

4. Comparative Analysis

- Built Top 10 Countries by GDP (Bar Chart):
    - Sorted by GDP descending
    - Highlighted dominant economic regions
- Used data labels and axis scaling for clarity

5. Categorical Analysis

- Used donut chart for GDP growth categories:
    - Low, Medium, High growth
- Applied custom color coding:
    - High → Green (growth)
    - Medium → Blue (stable)
    - Low → Red (risk)

6. Interactivity Features

a. Slicers:

- Country filter → dynamic country-level analysis
- Year range slider → time-based filtering

b. Drill-through:

- Created Country Detail Page
- Enabled deep analysis by right-clicking any country

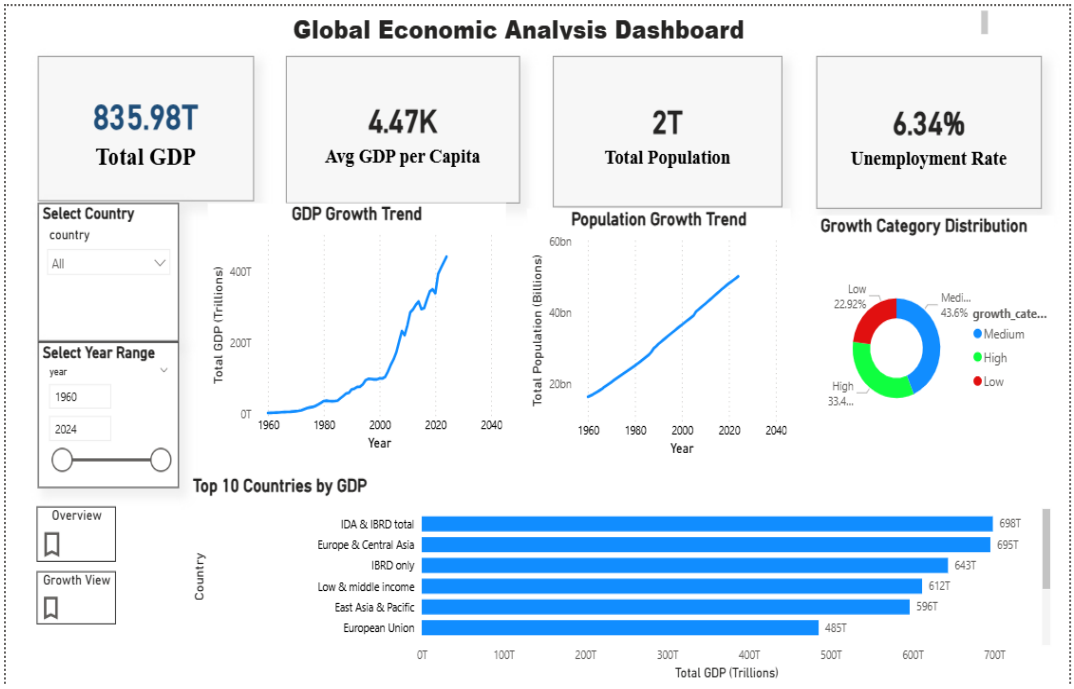
- Displays:
  - GDP trend
  - Population trend
  - Unemployment trend

c. Bookmarks:

- Created multiple views:
  - Overview → full dashboard
  - Growth View → focused analysis
- Added buttons for navigation
- Improved user experience and storytelling

d. Dashboard Design & UX

- Used clean layout with proper spacing and alignment
- Applied minimal color palette with one accent color
- Added titles and labels for clarity
- Ensured readability and non-cluttered design



Key Insights from Dashboard

1. Global GDP Trend Insight

- GDP shows sharp acceleration after 2000, indicating rapid economic expansion
- Growth is not linear that suggests impact of globalization and industrial growth

2. Population vs GDP Insight

- Population growth is steady and linear, while GDP growth is exponential
- Indicates that economic growth is driven by productivity, not just population increase

3. Unemployment Insight

- Unemployment remains relatively stable (~5–7%) over decades
- Shows resilience of global labor markets despite economic fluctuations

4. Economic Concentration Insight

- Top regions (Europe, East Asia, etc.) dominate GDP contribution
- Indicates economic imbalance and concentration of wealth

5. Growth Category Insight

- Majority of data falls under Medium growth (44%) that indicates global economic stability
- High growth (33%) that is expansion phases
- Low growth (23%) means fewer downturn periods

6. Country-Level Insight (Drill-through)

- Some countries show:
    - High population but lower GDP indicates low productivity
    - High GDP per capita but smaller population indicates strong economy

- Helps identify developed vs developing economies

Dashboard Summary

The dashboard transforms complex economic data into an interactive analytical tool. It highlights how global GDP has accelerated significantly over time, while population growth remains steady. This indicates that economic progress is driven more by efficiency and productivity rather than population size.

Through slicers, drill-through, and bookmarks, users can explore both global trends and country-level insights, enabling deeper understanding and data-driven decision making

Recommendations

1. Focus on Productivity-Driven Growth (Not Just Population Expansion)

- The analysis shows GDP growth is exponential, while population growth is linear
- Governments should invest in technology, automation, and skill development to increase output per individual
- Improving GDP per capita should be prioritized over just increasing total GDP
- This ensures sustainable and high-quality economic growth

2. Strengthen Employment Generation in High-Growth Economies

- Despite strong GDP growth, unemployment shows fluctuations, indicating uneven job distribution
- Policies should focus on job creation in emerging sectors (technology, services, manufacturing)
- Encourage entrepreneurship and MSMEs to absorb workforce efficiently
- Align education systems with industry demands to reduce skill mismatch

3. Reduce Economic Imbalance Across Regions

- Dashboard shows GDP is concentrated in a few dominant regions, highlighting inequality
- Governments and global institutions should promote:
    - Investment in low and medium-income regions
    - Infrastructure development
    - Trade expansion
- This will ensure more balanced global economic growth

4. Use Data-Driven Monitoring for Long-Term Planning

- Trends show varying growth patterns across countries and time
- Policymakers should adopt data-driven dashboards (like this) for:
    - Continuous monitoring of GDP, population, and employment
    - Early identification of economic slowdowns
- This enables proactive decision-making rather than reactive policies

TASK-6 : Video Presentation

Link : https://www.loom.com/share/99b2ff8a66994d44b6d39bcf3a235cc3

Conclusion

This project successfully transformed raw World Bank data into meaningful and interactive business insights using Python and Power BI. Through data collection, cleaning, exploratory analysis, and feature engineering, key economic indicators such as GDP, population, and unemployment were analyzed to understand global economic trends.

The interactive dashboard enabled dynamic exploration of data through slicers, drill-through, and bookmarks, allowing both global and country-level analysis. The findings highlight that economic growth is primarily driven by productivity improvements rather than population size, while also revealing regional imbalances and stable unemployment patterns.

Overall, the project demonstrates how raw web data can be converted into decision-ready intelligence, supporting data-driven strategies for sustainable and balanced economic development.